



How to achieve a 7 Star NatHERS Rating in a Class 1a Building

A Quick Guide on how Kingspan's Insulation Products can help achieve
7 Star Compliance in Victoria.



Introduction

Background and Purpose

Kingspan Insulation has produced this document as a simple guide to achieving a NatHERS 7 Star rating under the 2022 NCC, which is now in use within the states of New South Wales and Victoria.

In Australia, buildings are directly responsible for around 18% of carbon emissions, largely from heating or cooling energy demands. Making our buildings more thermally efficient is one of the most important and most straightforward steps we can take to reduce carbon emissions and help mitigate climate change. A key part of achieving this aim is being driven by changes to the National Construction Code (NCC).

The Nationwide House Energy Rating Scheme – NatHERS – is the framework that defines the energy ratings for new dwellings.

NatHERS drives energy efficiency improvements for Australia’s residential buildings by nationally standardising the energy rating assessment tools used to rate dwellings.

NatHERS’ 10-Star rating system estimates a home’s thermal performance – in essence, its heating and cooling needs. Information about a home’s design, energy performance, construction materials and climate zone all contribute to its star rating. NatHERS’ rating tools use calculations following

rigorous scientific research carried out by Australia’s National Science Agency, CSIRO.

Following the 2022 update to the National Construction Code (NCC), the minimum required NatHERS rating increased from 6 to 7 stars in some regions. The NCC 2022 version also includes more recent and more accurate weather data updates to the climate files used to calculate star ratings for NatHERS. The new files, which relate to 69 Australian Climate Zones, now include data from 1990 to the end of 2015, replacing data gathered from 1970 to 2005.

Kingspan have undertaken in-depth testing to measure the effect of increased insulation in sample single-storey and double-storey homes. It specifically concentrates on the parts that are relevant to building fabric insulation, whilst showing how compliance can be achieved using Kingspan’s insulation products for walls and floors.

This document specifically looks at the state of Victoria and Climate Zone 21.

Overview of Models

For both house models, the wall, roof and window frame colours have been considered as medium. They have assumed solar panels on the side of the roof, which faces North, but the front of the house has a west facing orientation. IC4 Rated Downlights have been used throughout. The Ceiling Perimeter Insulated to R2.5 as per AS 3999:2015.

Single-Storey Home



The example is a four bedroom single-storey home with a total floor area of 235.17m². This comprises 184.43m² of ground floor space, 25.72m² garage, 4.68m² porch and 10.34m² of alfresco.

Double-Storey Home



The example is a four bedroom double-storey home with a total floor area of 384.70m². This comprises 168.06m² of ground floor, 36.59m² garage, 3.93m² porch and 176.12m² of first floor space.

7 Star Compliance

Single-Storey Home

Timber Frame Buildings – Climate Zone 21 (Melbourne)

The table below provides guidance on how to achieve 7 Star compliance for a timber frame single-storey home by increasing insulation.

Ground Floor	Windows	Sliding Doors	External Wall Insulation	Ceiling Insulation	Garage Ceiling Insulation	Roof Insulation Blanket	Laundry Perimeter Wall Insulation (within the studs)
Concrete Slab on Ground	Triple Glazed Low-E	Triple Glazed Low-E	R2.7 + 25mm Kingspan Kooltherm® K12 Framing Board*	R7.0	None Required	None Required	R2.7
Concrete Slab on Ground	Triple Glazed Low-E	Triple Glazed Low-E	80 mm Kingspan Kooltherm K12 Framing Board	R7.0	None Required	None Required	80 mm Kingspan Kooltherm K12 Framing Board
Wafflepod	Double Glazed	Double Glazed	R2.7 + 25mm Kingspan Kooltherm K12 Framing Board*	R7.0	R7.0	None Required	R2.7
Wafflepod	Double Glazed	Double Glazed	80 mm Kingspan Kooltherm K12 Framing Board	R7.0	R7.0	None Required	80 mm Kingspan Kooltherm K12 Framing Board
Concrete Slab on Ground + Kingspan Kooltherm® K3 Floorboard (50mm) + Kingspan GreenGuard® GG350 (40 mm)	Double Glazed	Double Glazed	R2.7	R6.0	None Required	None Required	None Required
Concrete Slab on Ground + Kingspan Kooltherm K3 Floorboard (50mm) + Kingspan GreenGuard GG350 (40 mm)	Double Glazed	Single Glazed	R2.7 + 25mm Kingspan Kooltherm K12 Framing Board*	R7.0	None Required	None Required	R2.7
Concrete Slab on Ground + Kingspan Kooltherm K3 Floorboard (50mm) + Kingspan GreenGuard GG350 (40 mm)	Double Glazed	Single Glazed	80 mm Kingspan Kooltherm K12 Framing Board	R7.0	R7.0	None Required	80 mm Kingspan Kooltherm K12 Framing Board
* Added to the frame internally or externally							

Table 1: Achieving 7 Star Compliance in an example Timber Frame Single-Storey Home.

Steel Frame Buildings – Climate Zone 21 (Melbourne)

The table below provides guidance on how to achieve 7 Star compliance for a steel frame single-storey home by increasing insulation.

Ground Floor	Windows	Sliding Doors	External Wall Insulation	Internal Wall Insulation	Ceiling Insulation	Garage Ceiling Insulation	Roof Insulation Blanket	Laundry Perimeter Wall Insulation (within the studs)
Concrete Slab on Ground	Triple Glazed Low-E	Triple Glazed Low-E	R2.2 + 25mm Kingspan Kooltherm K12 Framing Board*	None Required	R7.0	R7.0	R1.3	R2.2
Wafflepod	Double Glazed	Double Glazed Low-E	R2.2 + 25mm Kingspan Kooltherm K12 Framing Board*	None Required	R7.0	R7.0	R1.3	R2.2
Concrete Slab on Ground + Kingspan Kooltherm K3 Floorboard (50mm) + Kingspan GreenGuard GG350 (40 mm)	Double Glazed	Double Glazed	R2.2	R2.2	R7.0	R7.0	R1.3	R2.2
Concrete Slab on Ground + Kingspan Kooltherm K3 Floorboard (50mm) + Kingspan GreenGuard GG350 (40 mm)	Double Glazed	Double Glazed	R2.2 + 25mm Kingspan Kooltherm K12 Framing Board*	None Required	R6.0	R6.0	None Required	R2.2
* Added to the frame internally or externally								

Table 2: Achieving 7 Star Compliance in an example Steel Frame Single-Storey Home.

7 Star Compliance

Double-Storey Home

Timber Frame Buildings – Climate Zone 21 (Melbourne)

The table below provides guidance on how to achieve 7 Star compliance for a double-storey home by increasing insulation.

Ground Floor	Windows	Sliding Doors	External Wall Insulation – Ground Floor	External Wall Insulation – First Floor	Ceiling Insulation	Roof Insulation Blanket	Laundry Perimeter Wall Insulation (within the studs)	Bathroom Perimeter Wall Insulation (within the studs)	W/C Perimeter Wall Insulation (within the studs)
Concrete Slab on Ground (uninsulated)	Double Glazed Low-E	Double Glazed Low-E	R2.5 + 25mm Kingspan Kooltherm K12 Framing Board*	R2.5 + 25mm Kingspan Kooltherm K12 Framing Board*	R6.0	None Required	R2.5	R2.5	R2.5
Concrete Slab on Ground (uninsulated)	Double Glazed Low-E	Double Glazed	80 mm Kingspan Kooltherm K12 Framing Board	80 mm Kingspan Kooltherm K12 Framing Board	R7.0	None Required	80 mm Kingspan Kooltherm K12 Framing Board	80 mm Kingspan Kooltherm K12 Framing Board	80 mm Kingspan Kooltherm K12 Framing Board
Wafflepod	Double Glazed Low-E	Double Glazed	R2.5 + 25mm Kingspan Kooltherm K12 Framing Board*	R2.5 + 25mm Kingspan Kooltherm K12 Framing Board*	R4.0	None Required	R2.5	None Required	None Required
Wafflepod	Double Glazed	Double Glazed	80 mm Kingspan Kooltherm K12 Framing Board	80 mm Kingspan Kooltherm K12 Framing Board	R7.0	R1.3	80 mm Kingspan Kooltherm K12 Framing Board	80 mm Kingspan Kooltherm K12 Framing Board	80 mm Kingspan Kooltherm K12 Framing Board
Concrete Slab on Ground + Kingspan Kooltherm K3 Floorboard (50mm) + Kingspan GreenGuard GG350 (40 mm)	Double Glazed	Double Glazed	R2.7	R2.7	R7.0	None Required	R2.7	R2.7	R2.7
Concrete Slab on Ground + Kingspan Kooltherm K3 Floorboard (50mm) + Kingspan GreenGuard GG350 (40 mm)	Double Glazed	Double Glazed	R2.7 + 25mm Kingspan Kooltherm K12 Framing Board*	R2.7 + 25mm Kingspan Kooltherm K12 Framing Board*	R7.0	None Required	None Required	None Required	None Required
Concrete Slab on Ground + Kingspan Kooltherm K3 Floorboard (50mm) + Kingspan GreenGuard GG350 (40 mm)	Double Glazed	Double Glazed	80 mm Kingspan Kooltherm K12 Framing Board	80 mm Kingspan Kooltherm K12 Framing Board	R5.0	None Required	80 mm Kingspan Kooltherm K12 Framing Board	None Required	None Required

* Added to the frame internally or externally

Table 3: Achieving 7 Star Compliance in an example Timber Frame Double-Storey Home.

7 Star Compliance

Double-Storey Home

Steel Frame Buildings – Climate Zone 21 (Melbourne)

The table below provides guidance on how to achieve 7 Star compliance for a double-storey home by increasing insulation.

Ground Floor	Windows	Sliding Doors	External Wall Insulation – Ground Floor	External Wall Insulation – First Floor	Ceiling Insulation	Roof Insulation Blanket	Laundry Perimeter Wall Insulation (within the studs)	Bathroom Perimeter Wall Insulation (within the studs)	W/C Perimeter Wall Insulation (within the studs)
Concrete Slab on Ground (uninsulated)	Double Glazed Low-E	Double Glazed Low-E	R2.2 + 25mm Kingspan Kooltherm K12 Framing Board*	R2.2 + 25mm Kingspan Kooltherm K12 Framing Board*	R7.0	R1.3	R2.2	R2.2	R2.2
Wafflepod	Double Glazed Low-E	Double Glazed	R2.2 + 25mm Kingspan Kooltherm K12 Framing Board*	R2.2 + 25mm Kingspan Kooltherm K12 Framing Board*	R6.0	None Required	R2.2	R2.2	R2.2
Concrete Slab on Ground + Kingspan Kooltherm K3 Floorboard (50mm) + Kingspan GreenGuard GG350 (40 mm)	Double Glazed Low-E	Double Glazed	R2.2	R2.2	R5.0	None Required	R2.2	None Required	None Required
Concrete Slab on Ground + Kingspan Kooltherm K3 Floorboard (50mm) + Kingspan GreenGuard GG350 (40 mm)	Double Glazed	Double Glazed	R2.2 + 25mm Kingspan Kooltherm K12 Framing Board*	R2.2 + 25mm Kingspan Kooltherm K12 Framing Board*	R7.0	R1.3	R2.2	None Required	None Required
* Added to the frame internally or externally									

Table 4: Achieving 7 Star Compliance in an example Steel Frame Double-Storey Home.

Insulation Solutions for Walls

External Insulation

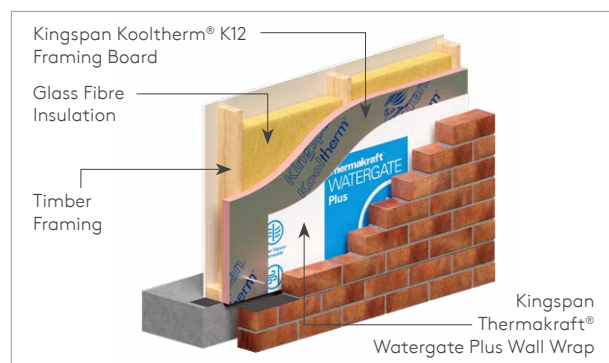


Figure 1. Externally lined Continuous Insulation for Timber Frame.

Build up to achieve R2.7 within the stud with 25 mm of Kingspan Kooltherm K12 Framing Board on the outside (Refer Figure 1)

Wall Build Up

Stud Depth (mm)	Between studs insulation – Glass Fibre Insulation (mm)	External Insulation – Kingspan Kooltherm K12 Framing Board (mm)
90	90 (R2.7)	25 (R1.1)

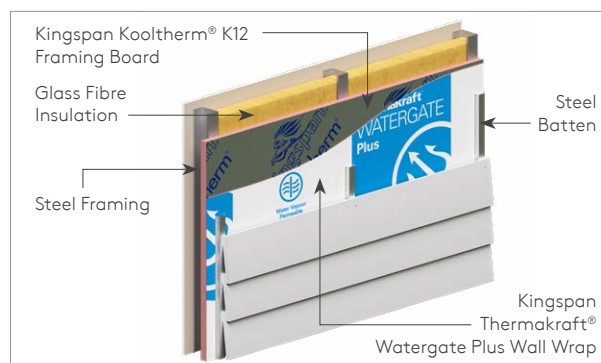


Figure 2. Externally lined Continuous Insulation for Steel Frame.

Build up to achieve R2.2 within the stud with 25 mm of Kingspan Kooltherm K12 Framing Board on the outside (Refer Figure 2)

Wall Build Up

Stud Depth (mm)	Between studs insulation – Glass Fibre Insulation (mm)	External Insulation – Kingspan Kooltherm K12 Framing Board (mm)
76	75 (R2.2)	25 (R1.1)

Internal Insulation

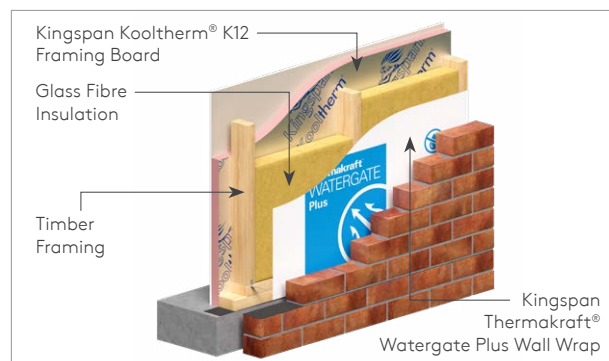


Figure 3. Internally lined Continuous Insulation for Timber Frame.

Build up to achieve R2.7 within the stud with 25 mm of Kingspan Kooltherm K12 Framing Board on the inside (Refer Figure 3)

Wall Build Up

Stud Depth (mm)	Between studs insulation – Glass Fibre Insulation (mm)	Internal Insulation – Kingspan Kooltherm K12 Framing Board (mm)
90	90 (R2.7)	25 (R1.1)

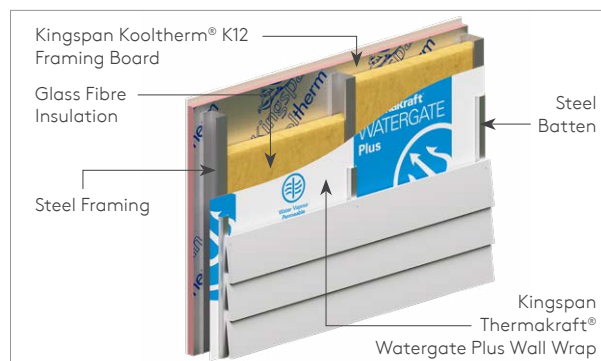


Figure 4. Internally lined Continuous Insulation for Steel Frame.

Build up to achieve R2.2 within the stud with 25 mm of Kingspan Kooltherm K12 Framing Board on the inside (Refer Figure 4)

Wall Build Up

Stud Depth (mm)	Between studs insulation – Glass Fibre Insulation (mm)	Internal Insulation – Kingspan Kooltherm K12 Framing Board (mm)
76	75 (R2.2)	25 (R1.1)

Insulation Solutions for Walls

Rigid Insulation In-Between Studs

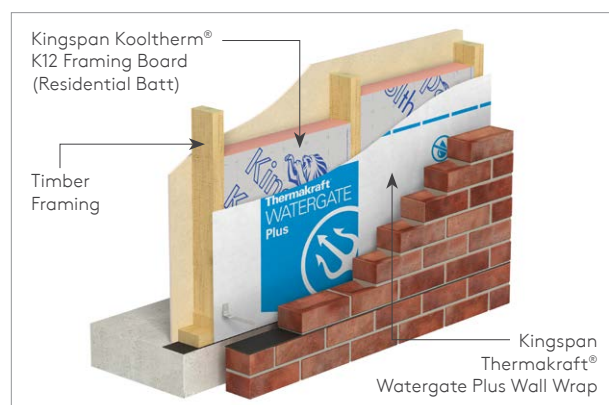


Figure 5. Rigid Board Insulation between Studs for Timber Frame.

Build up for Kingspan Kooltherm K12 Framing Board on the inside of the Timber Frame (Refer Figure 5)

Wall Build Up

Stud Depth (mm)	Between studs insulation – Kingspan Kooltherm K12 Framing Board (mm)
90	80 (R3.6)

Insulation Solutions for Floors

Underfloor and Slab Edge Insulation

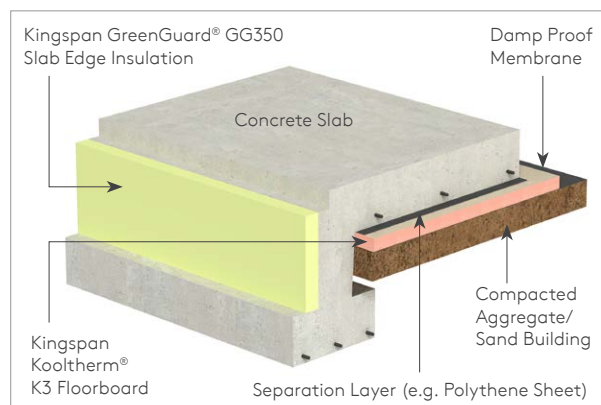


Figure 6. Underfloor and Slab Edge Insulation.

Concrete Slab on Ground + Kingspan Kooltherm K3 Floorboard 50 mm + Kingspan GreenGuard GG350 40 mm (Refer Figure 6)

Floor Build Up

Under slab insulation – Kingspan Kooltherm K3 Floorboard (mm)	Perimeter Insulation – Kingspan GreenGuard GG350 (mm)
50 (R2.3)	40 (R1.0)

Product Pack Sizes for Walls



Kingspan Kooltherm K12 Framing Board Pack Sizes

25 mm x 2400 mm x 1200 mm	12 x per pack
80 mm x 1200 mm x 403 mm	6 x per pack
80 mm x 1200 mm x 413 mm	6 x per pack
80 mm x 1200 mm x 553 mm	6 x per pack
80 mm x 1200 mm x 563 mm	6 x per pack



Thermakraft Watergate Plus Wall Wrap Pack Sizes

1370 mm wide x 36.5 m long	50m ² coverage*
1500 mm wide x 30 m long	45m ² coverage*
2740 mm wide x 30 m long	82m ² coverage*
3000 mm wide x 30 m long	90m ² coverage*

* Note:
m² is the roll size. For actual coverage, allow for laps and joins.

Product Pack Sizes for Floors



Kingspan Kooltherm K3 Floorboard Pack Sizes

25 mm x 2400 mm x 1200 mm	12 x per pack
50 mm x 2400 mm x 1200 mm	6 x per pack



Kingspan GreenGuard GG350 Slab Edge Insulation Pack Sizes

40 mm x 2400 mm x 300 mm	5 x per pack
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